

Design, Simulation and Experimental Validation of Silicon Strip Waveguides and Mach–Zehnder Interferometers

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Abstract:

Silicon photonics enables the integration of optical components on a single chip, offering compact, high-speed, and energy-efficient solutions for modern communication and sensing applications. This report presents the design, simulation, and experimental characterization of silicon photonic devices, beginning with the modal analysis of a silicon strip waveguide and extending to the performance evaluation of Mach–Zehnder Interferometers (MZIs). Key parameters including free spectral range (FSR), group index, and insertion loss are extracted from simulated and measured data. Corner and Monte Carlo analyses are also performed to assess the impact of fabrication variations, providing a comprehensive evaluation of device performance and reliability.

1. Introduction

Silicon photonics has become a leading platform for Photonic Integrated Circuits (PICs) due to its CMOS compatibility, compact footprint, and high-performance optical functionality. The design of photonic devices begins with understanding light propagation in silicon waveguides through TE and TM mode analysis. Building on this foundation, Mach–Zehnder Interferometers (MZIs) are studied because of their wide use in optical modulation, filtering, sensing, and signal processing. This report presents the simulation and experimental analysis of MZIs, including the extraction of FSR, group index, and insertion loss, followed by corner and Monte Carlo analyses to evaluate the effect of fabrication variations and verify the robustness of the design.

Waveguide characterization is performed to evaluate efficient confinement and guiding of light in a silicon waveguide. It determines key parameters such as the effective refractive index (n_{eff}), supported optical modes (TE/TM), propagation loss, group index, and polarization behavior. Since every photonic component is built using waveguides, accurate characterization is essential before designing more complex devices such as MZIs and ring resonators.

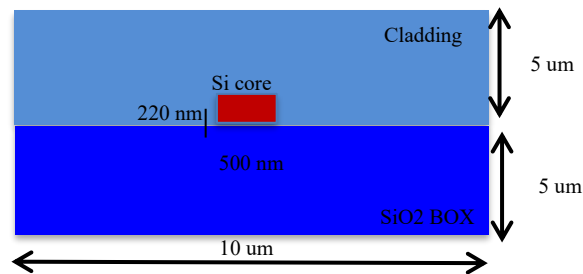


Fig. 1. Cross-section of SOI waveguide

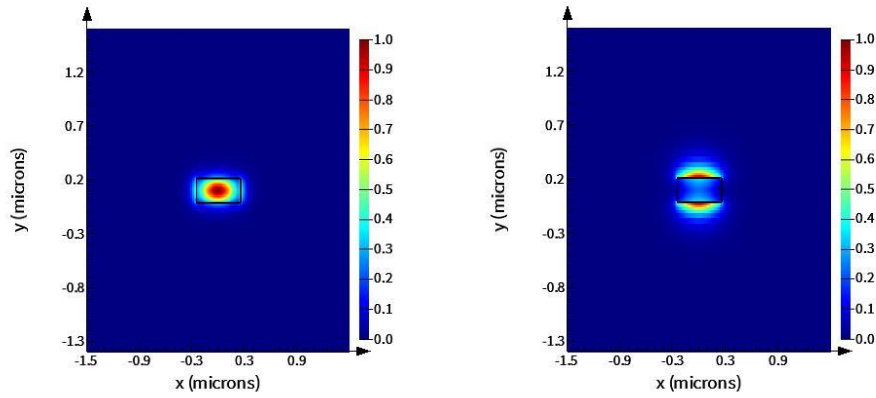


Fig. 2. Left image shows TE Mode Profile & Right image shows TM Mode Profile (Simulation)

The simulated TE mode showed strong optical confinement within the silicon core, while the TM mode exhibited greater field penetration into the cladding due to its field orientation. The effective indices obtained from simulation confirmed single-mode operation for the designed waveguide dimensions. These results establish the waveguide as a suitable building block for subsequent passive photonic devices such as directional couplers, Mach–Zehnder interferometers, and ring resonators.

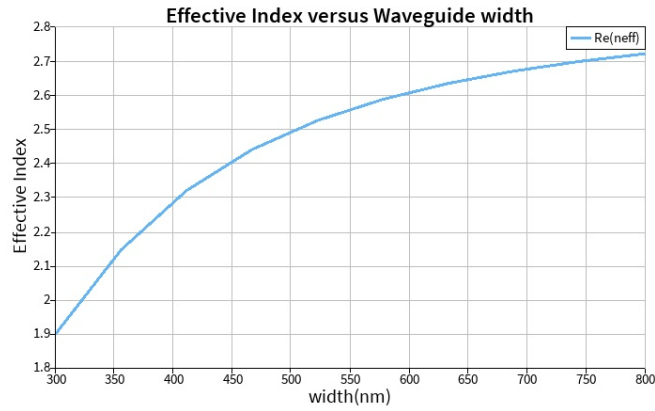


Fig. 3. Effective Index vs Waveguide Width

mode #	effective index	wavelength (μm)	loss (dB/cm)	group index	TE polarization fraction (Ex)	waveguide TE/TM fraction (%)	effective area (μm ²)
1	2.443942+2.138482e-09i	1.55	0.00075295	4.271157-4.000864e-09i	98	76.49 / 81.56	0.192259
2	1.775965+1.200584e-09i	1.55	0.00042272	3.786658+3.813983e-09i	4	68.99 / 88.7	0.349343
3	1.496220+5.939587e-10i	1.55	0.00020913	2.429066+4.774758e-09i	70	86.32 / 96.07	0.836407
4	1.409119+3.892398e-11i	1.55	1.0184e-05	1.520559-9.282654e-11i	100	98.52 / 96.4	5.22634
5	1.400459+2.375912e-12i	1.55	8.3655e-07	1.498709+1.064915e-11i	26	99.6 / 93.99	6.25275
6	1.395194+3.225690e-11i	1.55	1.1358e-05	1.570294-2.224429e-10i	0	95.45 / 95.28	4.39164
7	1.371959+1.298510e-10i	1.55	4.5720e-05	1.752672-1.412970e-09i	74	86.66 / 98.52	5.50746
8	1.369587+1.11913e-10i	1.55	3.9150e-05	1.739229+1.649810e-09i	14	98.97 / 81.22	5.36847
9	1.352800+4.180499e-12i	1.55	1.4719e-06	1.554561+1.015786e-11i	88	99.94 / 87.18	6.51738
10	1.326698+2.227223e-12i	1.55	7.8420e-07	1.577986+4.896691e-13i	1	97.78 / 86.49	4.27339

Fig. 4. Mode Effective Index Table

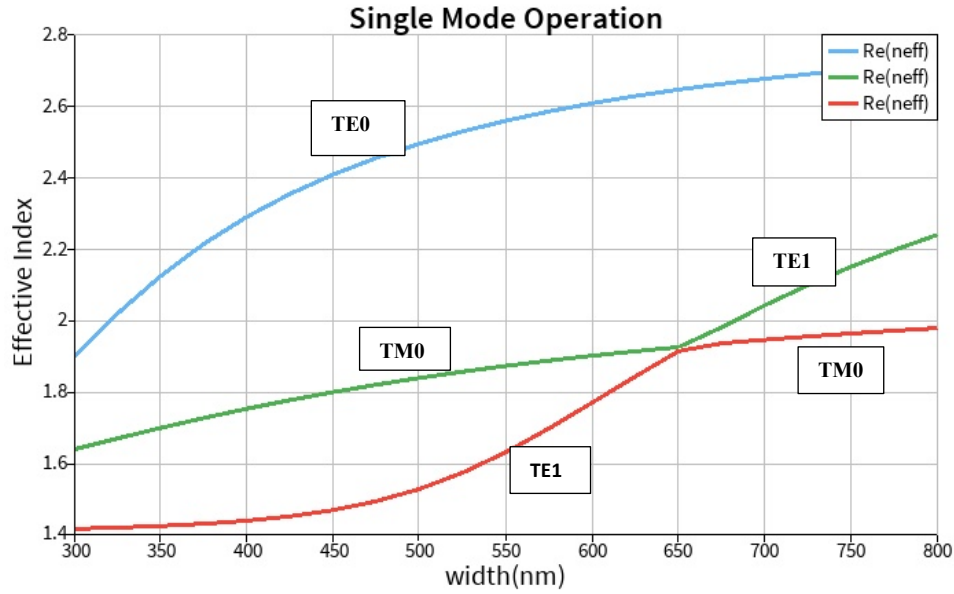


Fig. 5. Single-mode operation, Image also shows that how a point of intersection is given, where TE1 mode gets converted to TM0 and vice-versa (used for mode convertors).

A waveguide width of 500 nm is employed to achieve single-mode operation in the silicon waveguide. Single-mode propagation suppresses higher-order modes and reduces polarization-related effects, resulting in improved optical confinement and consistent device performance. Increasing the waveguide width beyond 500 nm allows higher-order modes to propagate. As a result, optical power can couple between the fundamental and higher-order modes, causing mode conversion and multimode propagation. Therefore, a 500 nm waveguide width is preferred to maintain stable single-mode operation.

Analysis of TE Waveguide Propagation Loss

Three TE waveguides (wg_TE_14_bends_340.737, wg_TE_14_bends_624.937, and wg_TE_14_bends_905.437) were designed with the same number of bends (14) but different waveguide lengths. Keeping the number of bends constant ensures that the bend loss is the same for all three devices. Therefore, any increase in the measured insertion loss is mainly due to the increase in waveguide length.

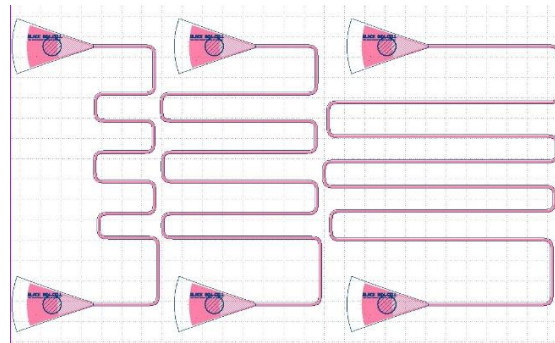


Fig. 6. GDS layout of the three waveguides

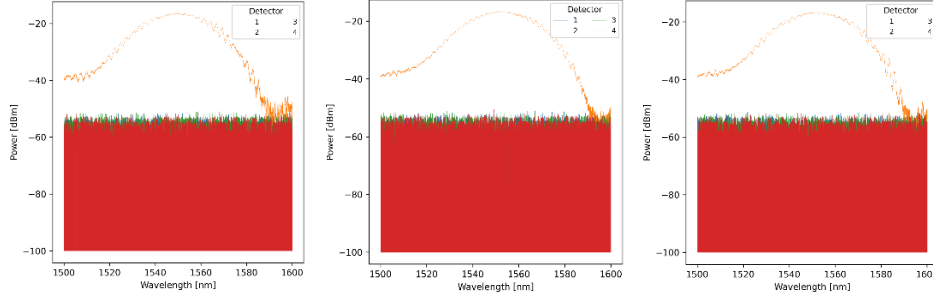


Fig. 7. Transmission spectra of the three waveguide devices measured after fabrication

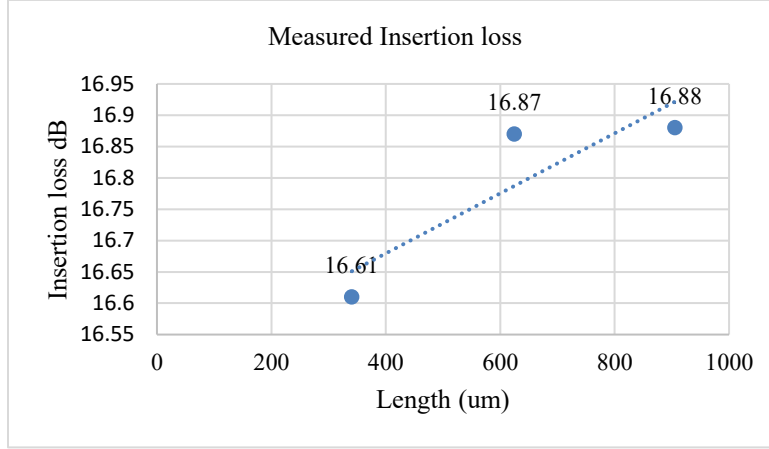


Fig. 8. Graph: Insertion Loss (dB) vs Waveguide Length with a best-fit straight line.

The insertion loss of each waveguide is measured and plotted against its length. A straight-line fit is applied to the data using the cut-back method. The slope of the line gives the intrinsic propagation loss (dB/cm), while the intercept represents the fixed losses, such as grating coupler and bend losses. Important for designing low-loss photonic integrated circuits.

S. No.	Device name	Length (um)	No. of bends	Measured IL	Simulated Loss
1	wg_TE_14_bends_340.737	340.737	14	-16.61	-5.01
2	wg_TE_14_bends_624.937	624.937	14	-16.87	-5.22
3	wg_TE_14_bends_905.437	905.437	14	-16.88	-5.40

Table I: Device name, length, number of bends, measured insertion loss and simulated IL.

Measured IL = Coupler Loss + Bend Loss + Propagation Loss + Other fixed Losses

Propagation loss = $Slope = (Loss_2 - Loss_1) / (Length_2 - Length_1)$

$$= (16.88 - 16.61) / (905.437 - 340.737)$$

$$= 0.27 / 564.7 = 4.78 \times 10^{-4} \text{ dB/um} = 4.78 \text{ dB/cm}$$

Intercept = -16.49 dB (constant losses)

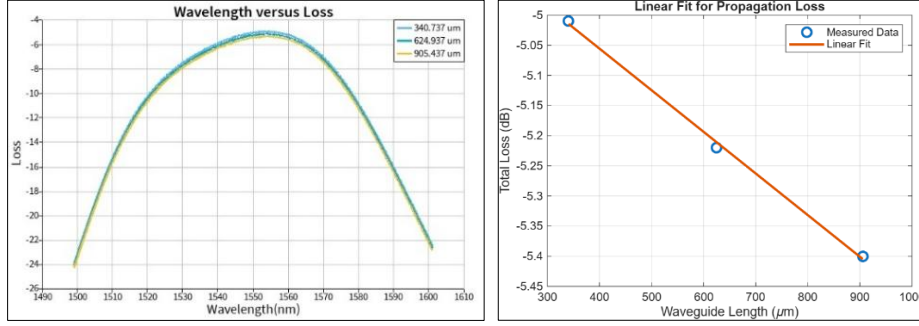


Fig. 9. Left image shows simulated loss versus wavelength & Right image shows simulated loss versus length of waveguide using matlab.

Parameter	Simulation	Measurement
Propagation Loss (dB/cm)	6.90	4.79
Y-intercept (dB)	-4.77	-16.49

Table II: Comparison of Simulated versus Measured results.

Propagation loss represents the optical power attenuated as light propagates through the waveguide due to material absorption, scattering, and sidewall roughness. In this work, the measured propagation loss was extracted using the cut-back method by plotting insertion loss against waveguide length and determining the slope of the best-fit line, while the simulated value was obtained from the INTERCONNECT PDK compact model. The measured insertion loss increased linearly with waveguide length, validating the cut-back method for propagation loss extraction.

The measured y-intercept (-16.49 dB) is significantly lower than the simulated value (-4.77 dB) because it includes additional fixed losses from grating couplers, fiber-to-chip coupling, alignment errors, and the experimental setup, which are not fully represented in the simulation. Although the simulated propagation loss (6.90 dB/cm) is higher than the measured value (4.79 dB/cm), this difference is likely due to the PDK compact model being generated from an earlier fabrication process with higher waveguide losses, whereas the current fabricated waveguides exhibit improved optical quality and lower propagation loss.

2. Device Design

A Mach-Zehnder Interferometer (MZI) is a fundamental photonic integrated device that operates on the principle of optical interference to modulate light intensity as a function of wavelength. It comprises an input splitter, two waveguide arms with a defined optical path length difference (ΔL), and an output combiner. As light propagates through the two arms, a phase difference is introduced due to the path length difference. Upon recombination, constructive or destructive interference occurs, producing the characteristic transmission spectrum of the MZI.

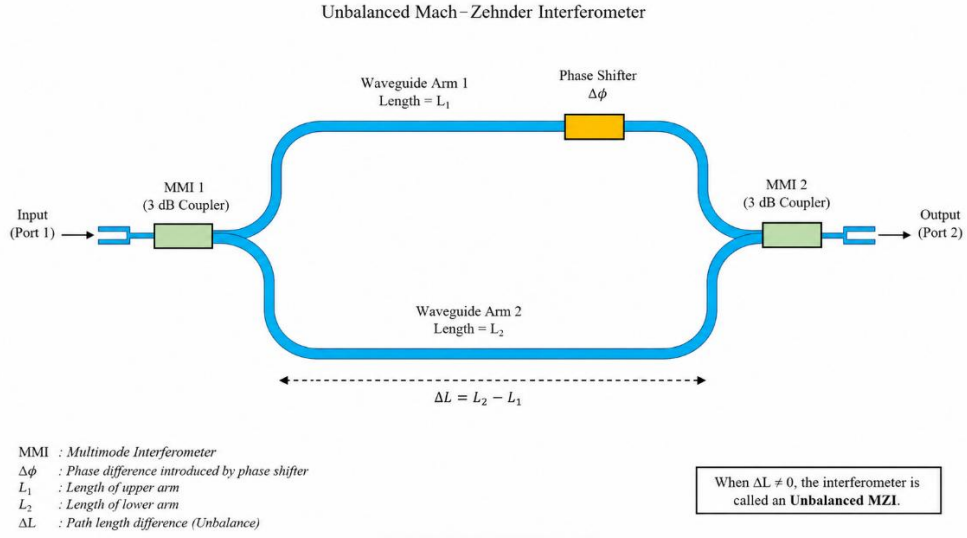


Fig. 10. Unbalanced MZI drawn for basic understanding

In this work, MZIs with different optical path length differences are designed and analyzed to investigate the effect of ΔL on device performance. Increasing ΔL increases the phase difference between the two arms, resulting in a reduction of the Free Spectral Range (FSR). Simulated and measured transmission spectra are compared to extract key performance parameters, including FSR, group index, and insertion loss. Furthermore, corner and Monte Carlo analyses are performed to evaluate the sensitivity of the MZI to fabrication variations and assess the robustness of the design.

Phase difference
$$\Delta\phi = \frac{2\pi}{\lambda} n_{eff} \Delta L$$

Output intensity
$$I = I_0 \cos^2(\Delta\phi/2)$$

Interference depends entirely on the phase difference accumulated in the two paths. It is used in optical modulators, filters, switches, sensors, and phase shifters.

Different MZI configurations are designed by varying the path length difference (ΔL) between the two arms. Device Parameters taken as waveguide platform of Silicon, waveguide width as 500 nm, waveguide thickness: 220 nm, bend radius: 10 μm , group index (n_g) as 4.05, operating wavelength range: 1500–1600 nm.

Device	Arm 1 Length	Arm 2 Length	ΔL
MZI-0	142.517	162.717	20.2
MZI-1	144.439	177.279	32.84
MZI-2	144.019	177.279	33.26
MZI-3	457.179	173.559	283.62

Table III: Path length variations taken for this analysis.

The output intensity of an MZI is governed by optical interference between two arms, increasing ΔL decreases FSR (denser fringes) and decreasing ΔL increases FSR (wider spacing).

3. Simulation Setup

The device was simulated using a photonic simulation tool (e.g., INTERCONNECT/ Lumerical). A broadband wavelength sweep from 1500 nm to 1600 nm was used. Simulation conditions were taken as input source of TE polarized broadband source, Wavelength sweep range from 1500 to 1600 nm, Y-branch is used, the output transmission spectra were recorded for each MZI configuration.

4. Simulation Results

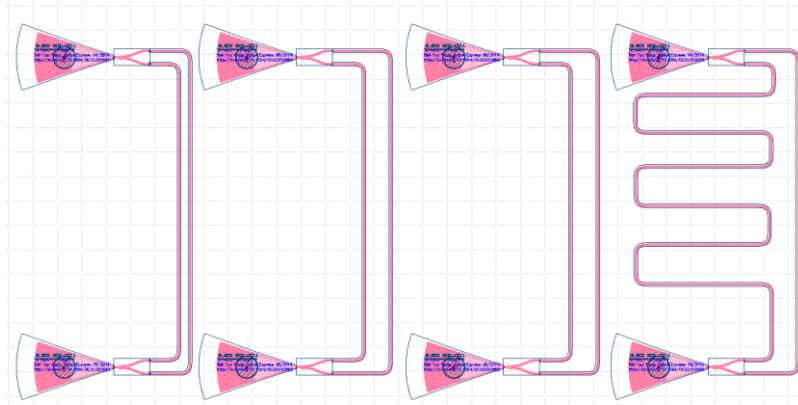


Fig. 11. MZI0, MZI1, MZI2, MZI3 in order from left to right as designed in GDS file

The simulated spectral response shows periodic interference fringes whose spacing depends on the path length difference ΔL . From this analysis we observed that MZI with larger ΔL shows closely spaced fringes (smaller FSR), also MZI with smaller ΔL shows widely spaced fringes (larger FSR). For high fringe visibility ideal conditions can be applied to the MZI.

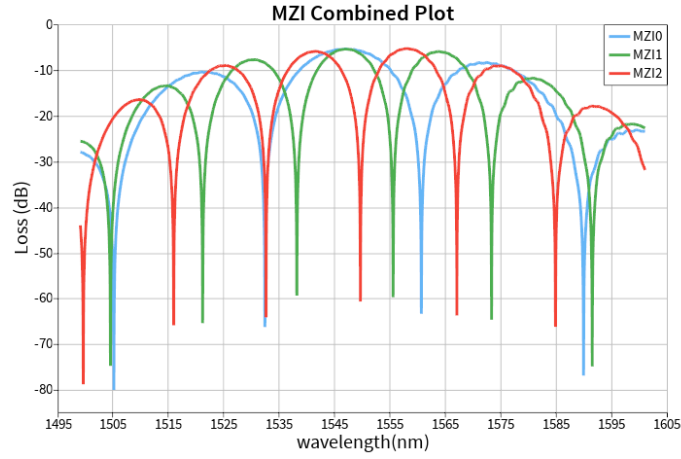


Fig. 12. Simulated Transmission vs Wavelength

5. Extracted Simulation Results

Device	ΔL	Simulated FSR (nm)	Group Index (ng)
MZI-1	32.84	16.376	4.47
MZI-2	33.26	16.4425	4.39
MZI-3	283.62	2.01	4.21

Table IV: Simulated FSR and Group Index for different length sing INTERCONNECT /Lumerical.

6. Experimental Measurement Results

The fabricated devices were characterized using an optical measurement setup with a tunable laser source and optical spectrum analyzer (OSA).

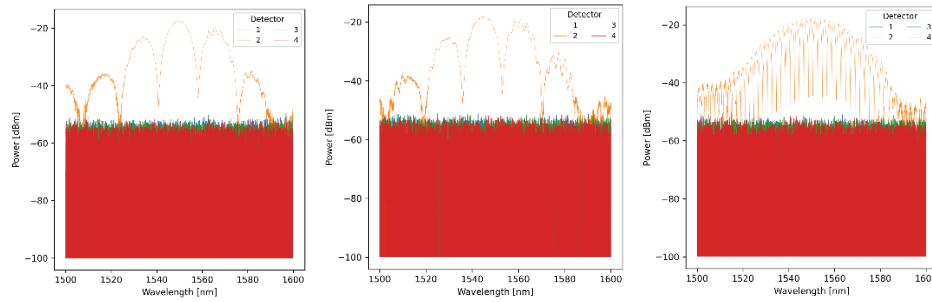


Fig. 13. Images obtained from measurement data of fabrication.

The fabricated devices were further characterized using matlab.

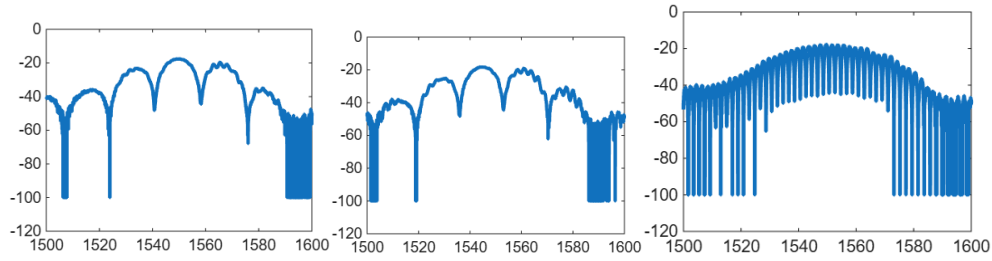


Fig. 14. Images produced using matlab importing the measurement data of fabrication.

Device	ΔL	Measured FSR (nm)	Group Index (ng)
MZI-1	32.84	16.87	4.34
MZI-2	33.26	16.19	4.46
MZI-3	283.62	1.88	4.51

Table V: Same devices were fabricated and measured for checking FSR and Group Index variations.

Observations:

Interference fringes are clearly observed in measured spectra. Slight deviation in FSR is observed compared to simulation. Reduced visibility is observed due to fabrication imperfections and losses.

6.1 Comparison of Simulation and Measurement

Device	Simulated FSR	Measured FSR	Error (%)
MZI-1	16.376	16.87	3.017
MZI-2	16.4425	16.19	1.54
MZI-3	2.01	1.88	6.47

Table VI: Comparison of simulation versus measurement results.

Differences between simulated and measured results may arise due to: Waveguide width and thickness variations during fabrication. Sidewall roughness leading to propagation loss. Coupler imbalance causing reduced interference visibility. Temperature fluctuations during measurement. Slight deviation in group index from designed value.

7. Manufacturing Variability

Corner Analysis (Process Variations): To evaluate fabrication robustness, corner analysis was performed by varying key parameters such as waveguide width, thickness, and refractive index. Manually performed 5 simulations in Lumerical MODE, for the 4 corners, and 1 nominal design. the minimum value for the group index was 4.17, the maximum value was 4.25, the minimum value for the FSR was 5.64 and maximum was 5.75.

The results indicate that waveguide width variations have a strong influence on effective index, leading to phase shift changes. However, the overall spectral behavior remains stable, indicating moderate robustness of the design.

8. Monte Carlo Analysis

Monte Carlo simulations were performed to evaluate the statistical variation of device performance under simultaneous random fabrication variations. The Monte Carlo results show that the MZI maintains stable performance under realistic fabrication variations. The distribution of FSR remains tightly centred around the nominal value, indicating good process tolerance.

9. Summary of the fabrication description:

Fabricated is performed at one or more of these: Applied Nanotools and Washington Nanofabrication Facility. The following are the process descriptions.

Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4\text{:H}_2\text{O}_2$) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

Washington Nanofabrication Facility (WNF) silicon photonics process

The devices were fabricated using 100 keV Electron Beam Lithography [1]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 μm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 °C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 μm exposure field size. The machine grid used for shape placement was 1 nm,

while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 $\mu\text{C}/\text{cm}^2$ was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using plasma enhanced chemical vapor deposition (PECVD) in an Oxford Plasmalab System 100 with a silane (SiH_4) flow of 13.0 sccm, nitrous oxide (N_2O) flow of 1000.0 sccm, high-purity nitrogen (N_2) flow of 500.0 sccm, pressure at 1400mT, high-frequency RF power of 120W, and a platen temperature of 350C. During deposition, chips rest directly on a silicon carrier wafer and are buffered by silicon pieces on all sides to aid uniformity.

Measurement description:

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

9. Discussion

The analysis demonstrates that the MZI spectral response is strongly dependent on the optical path length difference (ΔL). Simulation results closely match theoretical predictions, validating the design model. Measured results show slight deviations from simulation due to fabrication imperfections and non-ideal coupler behavior. Corner analysis confirms that the device is moderately sensitive to waveguide geometry variations, particularly width-induced effective index changes. Monte Carlo analysis further validates the robustness of the design by showing limited statistical spread in key performance metrics such as FSR and visibility.

Overall, the results confirm that the designed MZI structures are suitable for integrated photonic applications requiring stable interference behavior.

10. Conclusion

In this work, MZIs with varying path length differences were designed, simulated, fabricated, and analyzed. The study successfully demonstrates the dependence of spectral response on ΔL , with both simulated and measured results showing consistent trends.

Corner and Monte Carlo analyses confirm that the design is robust under realistic fabrication variations. The MZI structures exhibit stable interference behavior, making them suitable for photonic integrated circuit applications.

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